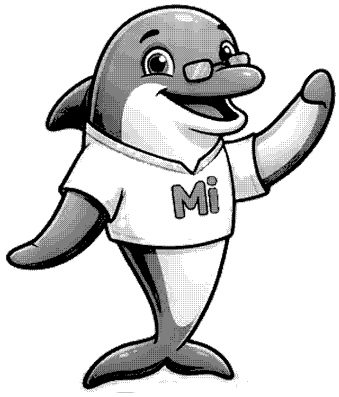
Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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Good to know

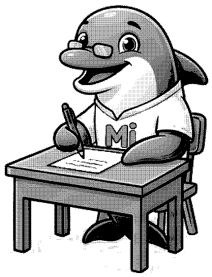
- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
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Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- The rulers in Unit 9 are printed at real size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Number sleuths (numbers to 10 000)	Not in this proof edition				☐
2	Cryptarithm crackers	Not in this proof edition				☐
3	Times table treasure	Not in this proof edition				☐
4	Model makers	6-7	8-9	10 and 12	11	☐
5	Remainder riddles	Not in this proof edition				☐
6	Fraction walls	Not in this proof edition				☐
7	Area architects	Not in this proof edition				☐
8	Angle and line detectives	Not in this proof edition				☐
9	Measure and money masters	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

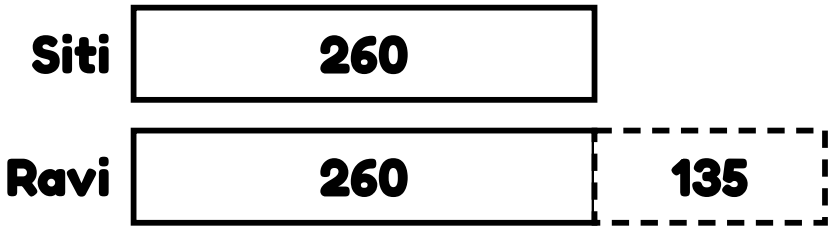
Learn it: Word problems with bar models

My goal. I can draw a bar model for a word problem. I can use it to decide what to add, subtract or divide.

A bar model is a picture of a word problem. Each bar is a length that shows an amount. Longer bar, greater amount.

Worked example

Siti has 260 beads. Ravi has 135 more beads than Siti. How many beads do they have in all?



1. **Draw a bar for Siti. Draw a longer bar for Ravi. The extra part is 135.**
2. **Ravi has 260 + 135 = 395 beads.**
3. **Both children: 260 + 395 = 655 beads.**

Answer: They have 655 beads in all.

How to draw a bar model

1. Draw one bar for each amount.
2. Write what you know on the bars. Put a ? on what you want to find.
3. Look at the bars. Then add, subtract or divide.

Your turn

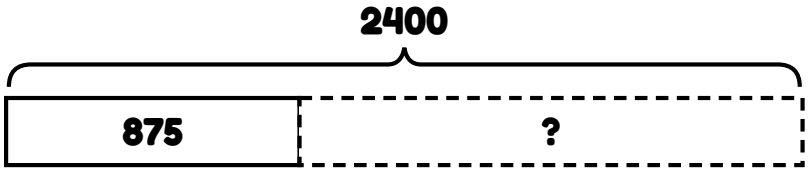
Answer at the back of the book.

Try **Ali has 480 stamps. Ben has 215 fewer stamps than Ali. How many stamps does Ben have?** ☐

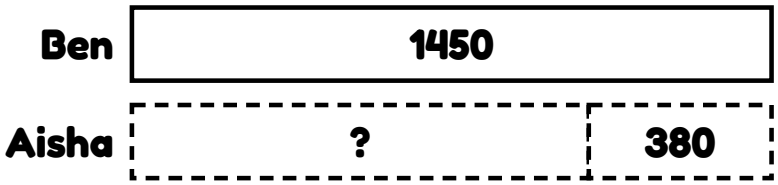
Start here

Do these first. Take your time.

1 A farm has 2400 eggs. 875 of them are brown. The rest are white. How many eggs are white? ☐



2 Ben has 1450 cards. Aisha has 380 fewer cards than Ben.



(a) Aisha has ☐ cards.

(b) They have ☐ cards in all.

3 A school sells 468 tickets on Friday, 539 on Saturday and 392 on Sunday.

(a) It sells ☐ tickets in all.

(b) It sells ☐ more tickets on Saturday than on Sunday.

4 Siti has 64 beads. Wei has 29 beads. Each child gets 12 more beads.

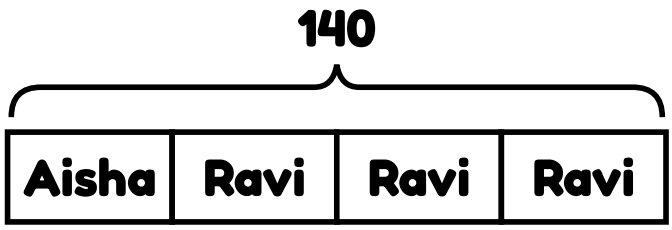
(a) Before, Siti has ☐ more beads than Wei.

(b) After, Siti has ☐ more beads than Wei.

Start here

Keep going. Check your work.

5 **Ravi and Aisha have 140 coins in all. Ravi has 3 times as many coins as Aisha.**



(a) Aisha has ☐ coins.

(b) Ravi has ☐ coins.

Working: _____

6 **Kumar and Wei have 90 stickers in all. Kumar has 18 more stickers than Wei.**

(a) Wei has ☐ stickers.

(b) Kumar has ☐ stickers.

Draw your model:

Working: _____

7 **Ravi's ribbon is 78 cm long. Mei's ribbon is 52 cm long. Ravi cuts a piece off his ribbon and gives it to Mei. Now both ribbons are the same length. How long is the piece?** ☐ **cm**

Draw your model:

Working: _____

Start here

Nearly there. You can do it.

8 **Ravi reads 85 pages on Monday. On Tuesday he reads 28 pages more than he does on Monday. On Wednesday he reads 17 pages fewer than on Tuesday.**

(a) He reads ☐ pages on Tuesday.

(b) He reads ☐ pages on Wednesday.

(c) He reads ☐ pages in the 3 days.

9 **Mei has $50. She buys a book for $18.40 and a pen for $6.75. How much money does she have left? $** ☐

10 **Wei has 3 times as many marbles as Aisha. Wei has 48 marbles.**

(a) Aisha has ☐ marbles.

(b) They have ☐ marbles in all.

Draw your model:

11 **Ali has 54 stickers. Ben has 18 fewer stickers than Ali. Ravi says Ben has 54 + 18 = 72 stickers.**

(a) Is Ravi right? Yes No

(b) Ben has ☐ stickers.

12 **Siti saves $12 each week for 7 weeks. Then she spends $35. How much money does she have left? $** ☐

Working: _____

Think harder

Hints are upside down. Try first!

1 **Kumar and Wei have 84 marbles in all. Kumar has 3 times as many marbles as Wei. Kumar gives some marbles to Wei. After that, each boy has the same number of marbles. How many marbles does Kumar give away?** ☐

Draw your model:

Working: _____

Hint. Find how many each has at first. Then how many each has later.

2 **Ali buys a pencil box and 2 pens for $14. Mei buys the same box and 5 pens for $23.**

(a) 1 pen costs $ ☐.

(b) The pencil box costs $ ☐.

Draw your model:

Working: _____

_____

Hint. Compare the 2 bills. What do the 3 extra pens cost?

3 **Ravi is 7 years old. His father is 35 years old. In how many years will his father be twice as old as Ravi?** ☐ **years**

Draw your model:

Working: _____

Hint. The age gap never changes. Draw a bar for each age.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Aisha, Kumar and Lily have 140 stickers in all. Kumar has 12 more stickers than Lily. Aisha has twice as many stickers as Kumar.**

(a) Lily has ☐ stickers. **(b)** Kumar has ☐ stickers.

(c) Aisha has ☐ stickers.

Draw your model:

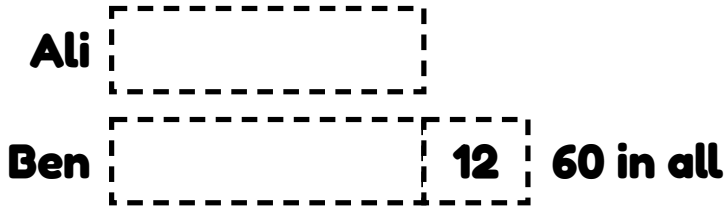
Hint. Draw Lily as 1 bar. Build Kumar and Aisha from that bar.

2 **Beyond Level 3: One amount changes.**
A shop has 4 times as many red balloons as blue balloons. It sells 18 red balloons. Now the shop has the same number of red and blue balloons. How many balloons did the shop have at first? ☐

Draw your model:

Hint. Draw blue as 1 box and red as 4 boxes. Selling 18 leaves 1 box.

3 **Look at the bar model. Make up a story that fits it. Then solve your story.**



My story: _____

_____

My answer: _____

Hint. Pick things that fit: stickers, coins, cards or beads.

✔ Check yourself

No hints. Try each one on your own.

1 **A library has 2000 books. 735 books are lent out. 640 books are being mended. How many books are on the shelves?** ☐

Working: _____

2 **Ravi has 560 beads. Siti has 135 more beads than Ravi. How many beads do they have in all?** ☐

Working: _____

3 **Siti and Kumar have 72 stamps in all. Siti has 5 times the number of stamps that Kumar has. How many stamps does Siti have?** ☐

Working: _____

4 **Two boxes hold 50 pens in all. Box A has 8 more pens than Box B. How many pens are in Box A?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths (numbers to 10 000)	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 3.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 4 · Model makers

Your turn
page 6

Answer: 265
Why: Ben has fewer, so subtract: 480 - 215 = 265. Check: 265 + 215 = 480.

START HERE

1
page 7

Answer: 1525
Why: The whole bar is 2400 eggs. One part is 875, so the white part is 2400 - 875 = 1525. Check: 875 + 1525 = 2400.

2
page 7

Answer: (a) 1070 (b) 2520
Why: (a) Aisha has fewer, so subtract: 1450 - 380 = 1070. (b) Together: 1450 + 1070 = 2520.

3
page 7

Answer: (a) 1399 (b) 147
Why: (a) 468 + 539 = 1007, and 1007 + 392 = 1399. (b) 539 - 392 = 147.

4
page 7

Answer: (a) 35 (b) 35
Why: Before: 64 - 29 = 35. After: Siti has 76 and Wei has 41, and 76 - 41 = 35. The gap stays the same because both children get the same number of beads.

5
page 8

Answer: (a) 35 (b) 105
Why: There are 4 equal boxes in all. 140 ÷ 4 = 35, so 1 box is 35. Aisha has 1 box: 35 coins. Ravi has 3 boxes: 3 × 35 = 105 coins.

6
page 8

Answer: (a) 36 (b) 54
Why: Take away the extra 18 from the total: 90 - 18 = 72. Share the 72 into 2 equal parts: 36 each. Wei has 36. Kumar has 36 + 18 = 54. Check: 36 + 54 = 90.

7
page 8

Answer: 13
Why: The gap is 78 - 52 = 26 cm. The piece closes the gap from both sides, so it is half of the gap: 26 ÷ 2 = 13 cm. Check: 78 - 13 = 65 and 52 + 13 = 65.

8
page 9

Answer: (a) 113 (b) 96 (c) 294
Why: (a) 85 + 28 = 113. (b) 113 - 17 = 96. (c) 85 + 113 + 96 = 294. Wednesday is compared with Tuesday, not with Monday.

9
page 9

Answer: 24.85
Why: First add what she spends: 18.40 + 6.75 = 25.15. Then 50.00 - 25.15 = 24.85. Check: 25.15 + 24.85 = 50.00.

10
page 9

Answer: (a) 16 (b) 64
Why: Wei has 3 boxes and Aisha has 1 box. 3 boxes are 48 marbles, so 1 box is 48 ÷ 3 = 16. Together they have 4 boxes: 4 × 16 = 64.

11
page 9

Answer: (a) No (b) 36
Why: Ben has fewer stickers than Ali, so Ben has less than 54. Ravi added instead of subtracting. The right answer is 54 - 18 = 36.

12 page 9

Answer: 49
Why: In 7 weeks she saves 7 × $12 = $84. Then 84 - 35 = 49. She has $49 left.

THINK HARDER

1 page 10

Answer: 21
Why: At first there are 4 equal boxes: 84 ÷ 4 = 21, so Wei has 21 and Kumar has 63. At the end each has 84 ÷ 2 = 42. Kumar gives 63 - 42 = 21 marbles. Check: 21 + 21 = 42.

2 page 10

Answer: (a) 3 (b) 8
Why: The second bill has 3 more pens and costs $23 - $14 = $9 more. So 3 pens cost $9 and 1 pen costs $3. The box costs $14 - 2 × $3 = $8. Check: 8 + 15 = 23.

3 page 10

Answer: 21
Why: The gap is 35 - 7 = 28 years and it never changes. When the father is twice as old, Ravi is 1 box and his father is 2 boxes, so 1 box = the gap = 28. Ravi is 28, which is 21 years from now. Check: 28 and 56.

GO FURTHER

1 page 11

Answer: (a) 26 (b) 38 (c) 76
Why: Lily is 1 box. Kumar is 1 box and 12. Aisha is 2 boxes and 24. In all: 4 boxes and 36 = 140. So 4 boxes = 104 and 1 box = 26. Lily 26, Kumar 38, Aisha 76. Check: 26 + 38 + 76 = 140.

2 page 11

Answer: 30
Why: Red is 4 boxes and blue is 1 box. After selling 18 red balloons, red is 1 box. So 3 boxes are 18, and 1 box is 6. At first there were 6 blue and 24 red: 30 balloons in all.

3 page 11

Answer: Answers vary. Example: Ali and Ben have 60 stickers in all. Ben has 12 more stickers than Ali. Ali has 24 stickers and Ben has 36 stickers.
Why: Check that the story says 60 in all and 12 more for Ben, and that the answer works. In the example, take away the 12: 60 - 12 = 48, and 48 ÷ 2 = 24. Ali has 24 and Ben has 24 + 12 = 36.

CHECK YOURSELF

1 page 12

Answer: 625
Why: 2000 - 735 = 1265. Then 1265 - 640 = 625. Check: 735 + 640 + 625 = 2000.

2 page 12

Answer: 1255
Why: Siti has 560 + 135 = 695 beads. Together: 560 + 695 = 1255.

3 page 12

Answer: 60
Why: There are 6 equal boxes in all. 72 ÷ 6 = 12, so 1 box is 12. Siti has 5 boxes: 5 × 12 = 60.

4 page 12

Answer: 29
Why: Take away the extra 8: 50 - 8 = 42. Share 42 into 2 equal parts: 21 each. Box B has 21 and Box A has 21 + 8 = 29.